THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS UNIT TEST SET A

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | RBSE Board Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures; use graph paper where a graph is asked.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. Using prime factorisation, HCF$(96, 404)$ is: 1

(a) 4 **(b)** 2 **(c)** 12 **(d)** 8

2. A student states: "The product of two irrational numbers is always irrational." The correct refutation is: 1

(a) The statement is correct **(b)** True only for square roots

(c) False, since $\sqrt{2} \times \sqrt{2} = 2$ is rational **(d)** Cannot be determined

3. Two students find the LCM of 8 and 9. Student A says 72; Student B says 1 because the numbers are co-prime. Then: 1

(a) Student B is correct **(b)** Student A is correct

(c) Both are correct **(d)** Both are wrong

4. The graph of $y = p(x)$ touches the x-axis at exactly one point and does not cross it. The number of zeroes of $p(x)$ is: 1

(a) 1 **(b)** 3 **(c)** 2 **(d)** 0

5. A student states: "A quadratic polynomial can have at most three zeroes." This statement is: 1

(a) Correct **(b)** Incorrect — it can have at most 2 zeroes

(c) Correct for some polynomials **(d)** Cannot be determined

6. The zeroes of $x^2 - 9$ are: 1

(a) $0, 9$ **(b)** $3, 3$ **(c)** $9, -9$ **(d)** $3, -3$

7. If $x = a,\ y = b$ is the solution of $x - y = 2$ and $x + y = 4$, then (a, b) is: 1

(a) $(2, 2)$ **(b)** $(1, 3)$ **(c)** $(3, 1)$ **(d)** $(4, 0)$

8. If the lines $3x + 2ky = 2$ and $2x + 5y + 1 = 0$ are parallel, then k is: 1

(a) $\frac{5}{4}$ **(b)** $\frac{15}{4}$ **(c)** $\frac{4}{15}$ **(d)** $\frac{3}{2}$

9. Which point lies on the line $2x + 3y = 12$? 1

(a) $(0, 3)$ **(b)** $(3, 2)$ **(c)** $(6, 1)$ **(d)** $(2, 3)$

10. Which of the following is **not** a quadratic equation? 1

(a) $(x - 3)(2x + 1) = x(x + 5)$ **(b)** $x^2 - 2x = (-2)(3 - x)$ **(c)** $(x + 1)^2 = 2(x - 3)$ **(d)** $(x + 1)^2 = x(x + 3)$

11. The roots of $ax^2 + bx + c = 0$ $(a \neq 0,\ b^2 - 4ac \geq 0)$ are given by: 1

(a) $\frac{-b \pm \sqrt{b^2 + 4ac}}{2a}$ **(b)** $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ **(c)** $\frac{-b \pm \sqrt{b^2 - 4ac}}{a}$ **(d)** $\frac{b \pm \sqrt{b^2 - 4ac}}{2a}$

12. A student states: "$x^2 - 6x + 9 = 0$ has two distinct real roots." The correct response is: 1

(a) Correct

(b) Incorrect; $D = 0$, so the roots are equal $(3, 3)$

(c) Correct only for $x > 0$

(d) Cannot be determined

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

13. Express 3825 as a product of its prime factors. **2**

14. Check whether 6^n can end with the digit 0 for any natural number n. **2**

15. If α and β are the zeroes of $x^2 - 6x + 8$, find the value of $\alpha^2 + \beta^2$. **2**

16. The difference between two numbers is 26 and one number is three times the other. Find them. **2**

17. A student says that the roots of $\sqrt{2}x^2 + 7x + 5\sqrt{2} = 0$ are $-\sqrt{2}$ and $-\frac{5}{\sqrt{2}}$. Verify. **2**

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

18. Find the largest number which divides 245 and 1029 leaving remainder 5 in each case. **3**

19. Verify that $5, -2$ and $\frac{1}{3}$ are the zeroes of $3x^3 - 10x^2 - 27x + 10$. Also verify that the sum of the zeroes $= -\frac{b}{a}$. **3**

20. 5 pencils and 7 pens together cost ₹50, whereas 7 pencils and 5 pens together cost ₹46. Find the cost of one pencil and one pen. **3**

21. The area of a rectangular field is 528 m². Its length is one metre more than twice its breadth. Find the length and breadth. **3**

22. Rahul and Priya jog around a circular track. Rahul takes 16 minutes and Priya takes 20 minutes for one round. They start together from the same point at 6:00 a.m. in the same direction. When will they next meet at the starting point? **3**

OR

An army contingent of 616 members is to march behind an army band of 32 members in a parade. The two groups are to march in the same number of columns. What is the maximum number of columns in which they can march? **3**

23. A fraction becomes $\frac{1}{3}$ when 1 is subtracted from the numerator, and it becomes $\frac{1}{4}$ when 8 is added to its denominator. Find the fraction. **3**

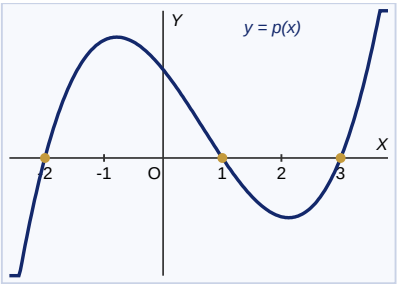
OR

Two years ago, a father was five times as old as his son. Two years later, his age will be 8 more than three times the age of the son. Find their present ages. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

24. **Case study:** The graph of a polynomial $y = p(x)$ is shown in the figure. **5**



(i) How many zeroes does $p(x)$ have? Write them.
(ii) What is the least possible degree of $p(x)$?
(iii) If the leading coefficient is 1, write $p(x)$ in standard form.
(iv) Verify that the sum of the zeroes $= -\frac{b}{a}$.

OR

If α and β are the zeroes of $x^2 - 5x + k$ such that $\alpha - \beta = 1$, find k and the zeroes. Also find a quadratic polynomial whose zeroes are 2α and 2β. **5**

25. The sum of the reciprocals of Rehman's ages (in years) 3 years ago and 5 years from now is $\frac{1}{3}$. Find his present age and verify. **5**

OR

In a class test, the sum of Shefali's marks in Mathematics and English is 30. Had she got 2 marks more in Mathematics and 3 marks less in English, the product of her marks would have been 210. Find her marks in the two subjects. **5**

END OF QUESTION PAPER — Total: 50 Marks